

DS 303: Statistical Foundations of Data Science

Course Code	DS 303
Course Name	Statistical Foundations of Data Science
L-T-P-C	3-0-0-3
Prerequisites	IC110-Engineering Mathematics, IC252-Data Science 2
Intended for	Discipline core for B. Tech. Data Science and Engineering, Discipline elective for B. Tech. Computer Science and Engineering, Electrical Engineering, Free elective for other B. Tech. disciplines
Approval	29th BoA (Mar 2019); 20th Senate (Apr 2019), in the B.Tech Data Science and Engineering curriculum

Course Contents

- **Module I:** Sample space, Sigma field, axiomatic definition of probability, conditional probability and independence, Bayes Rule. (2 lectures)
- **Review:** Random variables – discrete and continuous, probability mass function, probability density function, some standard (important) pdfs, independence, expectation, variance, conditional distribution, conditional expectation, covariance and correlation, Functions of random variables. (7 lectures)
- **Module III:** Probability generating function, moment generating function and characteristic functions – properties and applications. (3 lectures)
- **Module IV:** Convergence of random variables – basic results, inequalities (Markov and Chebyshev), law of large numbers (weak and strong), central limit theorem. (5 lectures)
- **Module V:** Random vectors and covariance and correlation matrix, Random processes – stationarity, WSS, Autocorrelation, cross correlation, power spectral density, Ergodicity. Wiener processes, Markov processes, Poisson Process. (8 lectures)
- **Sampling methods:** Inverse transforms sampling, Rejection sampling, adaptive rejection sampling, importance sampling, Markov chains and MCMC (8 lectures)
- **Graphical models:** ML and MAP estimation, directed and undirected models, Bayesian networks, CRF, Learning and Inference method (ML, MAP, Sampling) (9 lectures)

Text Books

1. Grimmett, Geoffrey, and David Stirzaker, **Probability and random processes**, Oxford university press, 2001.
2. Bishop, Christopher M., **Pattern recognition and machine learning**, Springer, 2006.

References

1. Ross, Sheldon., **A first course in probability**, Pearson, 2014.
2. Stark, Henry, and John William Woods, **Probability, statistics, and random processes for engineers**, Pearson, 2012.
3. Papoulis, Athanasios, and S. Unnikrishna Pillai, **Probability, random variables, and stochastic processes**, Tata McGraw-Hill Education, 2002.